AJAY KANNAN

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EDUCATION

Master of Science in Computer Science, Arizona State University, Tempe — Jan 2021 – May 2023

Relevant Coursework: Machine Learning, Deep Learning, Distributed Systems, Data Structures and Algorithms

Bachelor of Science in Computer Science, Anna University, Chennai — Aug 2014 – May 2019

PROFESSIONAL EXPERIENCE

Software Engineer | Pacific Gas & Electric, Santa Clara, CA — **Jul 2023 – Present**

- Designed and deployed a production data platform (Electric Productivity Tracker) processing large-scale utility datasets, improving data processing throughput by 20% and reducing system errors by 15% through backend automation, monitoring, and reliability engineering
- Engineered scalable, fault-tolerant data pipelines integrating heterogeneous sources (GIS, PostgreSQL, internal services), ensuring consistent data flow and low-latency access across distributed systems
- Built automated backend validation and testing infrastructure using Python and REST APIs, reducing deployment verification time by 50% through unit tests, integration tests, and end-to-end validation
- Implemented end-to-end data engineering workflows (ETL, validation, transformation), improving data integrity and reducing manual operational overhead through automated pipeline orchestration
- Automated extraction and processing of the Distribution Operations Toolset (DOT)—a daily-updated, macro-enabled Excel system containing nested map creation data—using Python and JavaScript to filter new maps, extract relevant information, and feed it into the map creation pipeline, eliminating manual data entry and reducing processing errors
- Unified three separate map creation processes into a single streamlined workflow, consolidating fragmented operations and improving overall efficiency, consistency, and maintainability across the team
- Applied ML techniques to improve anomaly detection in production systems, reducing false positives by 30% through model-assisted signal validation and statistical analysis
- Developed observability and monitoring systems (health checks, dashboards, alerting), reducing issue detection and resolution time by 50% through proactive alerting and centralized logging
- Collaborated with cross-functional teams through design reviews and code reviews, creating technical documentation including architecture diagrams, API specifications, and operational runbooks

Data Scientist | Sigtuple Inc., Bangalore, India — **Jul 2021 – Dec 2021**

- Applied image processing techniques to biological data using PyTorch, implementing Siamese Networks, GANs, and Neural Style Transfer for medical image analytics
- Utilized alpha blending techniques to recreate histology slide images, accurately representing biological particles and membranes
- Achieved 94% accuracy in image blending efficiency through innovative generative architectures
- Built production software in Python following object-oriented design with comprehensive unit and integration testing

Software Engineering | Microsoft, Hyderabad, India — **Summer 2017 & 2018**

- Developed ML algorithms for the FarmBeats agricultural platform using Python and PyTorch, identifying sensor data patterns to enhance crop yield predictions
- Built GIS-based heat-map generation system for FarmBeats, designing algorithms for real-time spatial visualization of multi-sensor data (soil moisture, temperature, humidity)
- Contributed to the Shopping on Cortana project, integrating Cortana with a shopping platform to improve user experience
- Developed backend services with Cassandra DB and C# as part of the Foundry Team in MS-IDC, building RESTful APIs for distributed data storage

TECHNICAL SKILLS

Programming Languages: Python (Expert), C++, Java, C#, JavaScript, SQL, Bash
ML/DL Frameworks: PyTorch, TensorFlow, Huggingface, Keras, NumPy, Scikit-learn, Pandas
LLM & AI Tools: LangChain, LLM Integration, Prompt Engineering, RAG Pipelines
Backend Development: REST APIs, FastAPI, Node.js, Express.js, Spring Boot, Microservices, GraphQL
Databases: PostgreSQL, MongoDB, MySQL, Cassandra, NoSQL
Cloud & DevOps: AWS (EC2, S3, Lambda), Azure, Docker, Kubernetes, Git, CI/CD
Computer Science: Data Structures, Algorithms, System Design, Distributed Systems, Object-Oriented Programming
Currently Learning: JAX, AWS Bedrock, Vector Databases (Pinecone), CUDA Programming

PROJECTS AND RESEARCH

Few-Shot Learning Research | Arizona State University **Aug 2022 – May 2023**

- Developed contrastive loss for Few Shot Learning in TCR-Epitope affinity prediction using PyTorch, improving accuracy by 30% through novel architectural improvements
- Implemented Siamese network architecture to evaluate efficiency of different K-values in few-shot learning, conducting comprehensive experiments and statistical analysis
- Presented research findings at ASU MORE symposium, demonstrating technical communication and quantitative reasoning skills

Gesture Recognition Systems | Solarillion Foundation **Aug 2015 – Dec 2018**

- Designed and developed both static and dynamic American Sign Language Gesture Recognition Systems achieving 94–97% accuracy using ML algorithms (Extra Trees, Random Forest, Ridge Classifier)
- Implemented advanced distance metric techniques to enhance recognition performance, contributing to accessible technology development for hearing-impaired users
- Built complete Python pipeline integrating sensor data acquisition, feature processing, model training, and real-time inference
- Led IoT research group, managing R&D projects and mentoring junior researchers as teaching assistant
- Published research in IEEE Global Internet of Things Summit 2017 and ArXiv

On-Device LLM Chatbot

- Built a voice-enabled LLM assistant using LM Studio and DeepSeek for local inference, exploring privacy-preserving conversational AI without cloud dependency
- Designed pipeline architecture handling speech-to-text, LLM inference, and text-to-speech components with error handling

Banking Chatbot with LangChain

- Built an LLM-powered banking chatbot using GPT models and LangChain, designing prompt workflows and intent handling for context-aware responses
- Implemented basic RAG pipeline to ground LLM responses in banking documentation, exploring vector embedding techniques
- Optimized NLP pipelines to handle diverse financial queries and improve response relevance

Twitter Data Analysis

- Built ML retrieval system using ANN and SVR achieving 98% accuracy in context-aware data retrieval
- Integrated streaming data pipelines with backend services and developed web application interface for real-time data visualization

PUBLICATIONS

- Few Shot Learning for TCR-Epitope Binding Affinity Prediction, MORE Symposium, ASU, 2023
- A. Kannan et al., "Low-cost static gesture recognition system using MEMS accelerometers," IEEE Global IoT Summit, 2017
- G. K. Gudur et al., "A Generic Multi-modal Dynamic Gesture Recognition System using Machine Learning," ArXiv abs/1809.05839, 2018